classiq for Quantum Solutions to the Paper Phase Transition

Cayman Quinn, Jaxon Kindred, Luke Papenfuss, Michael Hannesson

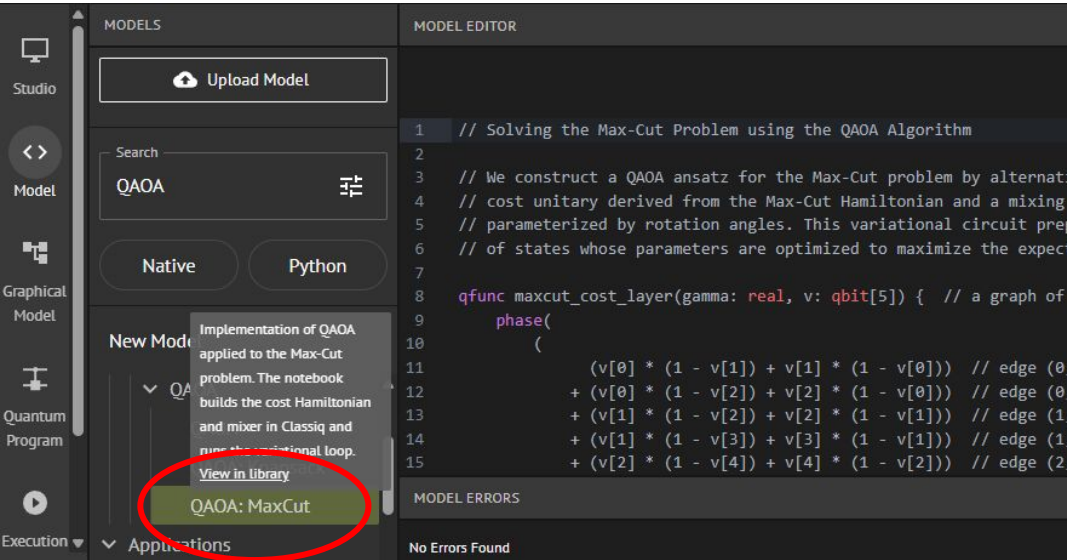




Using *classiq* to Learn and Visualize QAOA

classiq has many built-in workflows that assist in learning and visualizing quantum algorithms.

Once we understood our algorithms on a theoretical basis, **we needed a way for us to learn** how to make a quantum circuit for simulation.

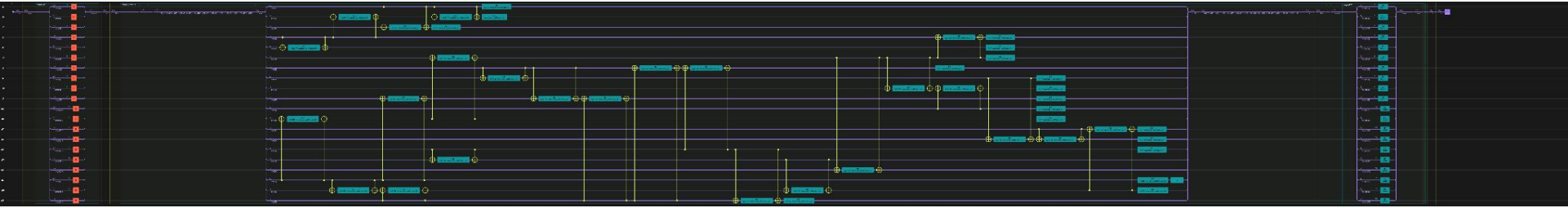


After looking around the infrastructure, we found an *example/pre-made* algorithm that **solved a simple case** of the quantum approximate optimization algorithm (QAOA) specific to *Max Cut*, similar to a standard algorithm used in Ising models.

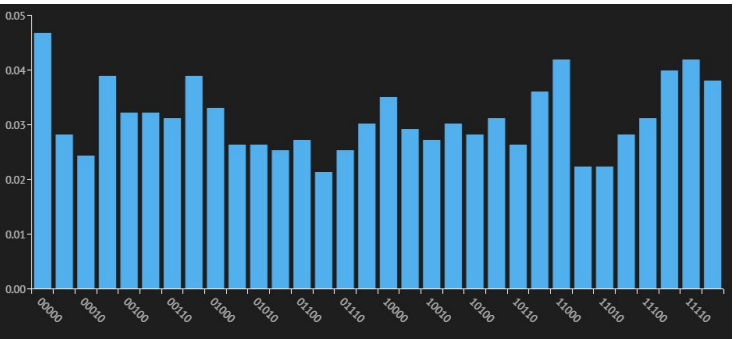
We took this base and **expanded upon it** to include the cost functions and weights of the *Vasion* Problem.

Using *classiq* to Learn and Visualize QAOA

From this, we were able to **synthesize** the python functions into a real quantum circuit.



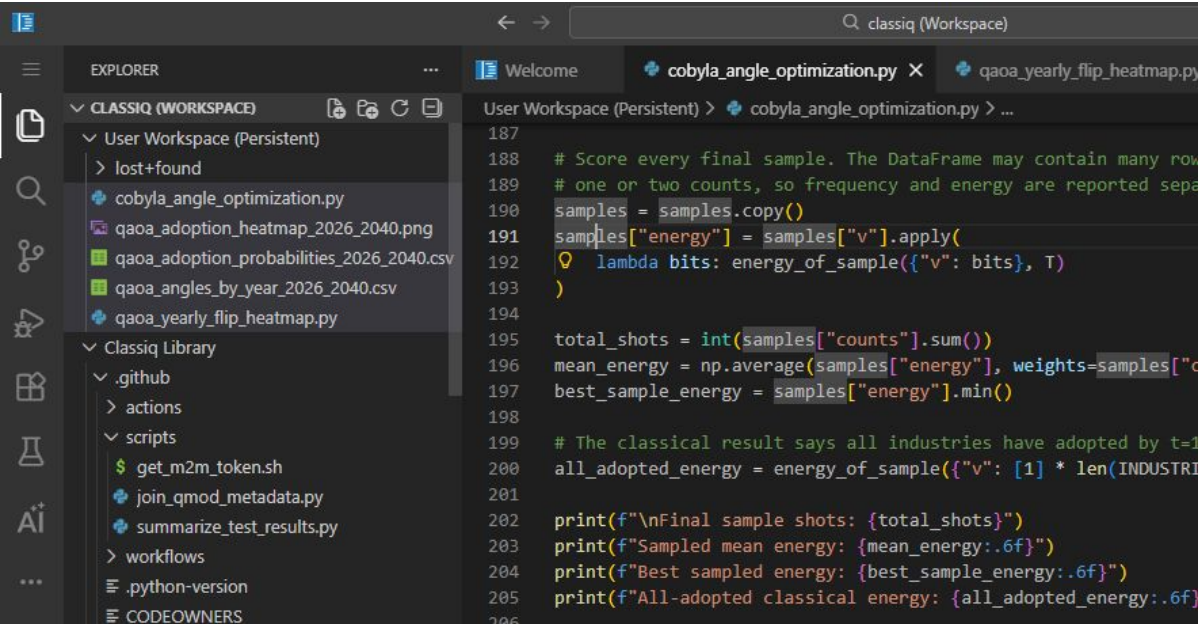
Running our algorithm on this initially produced mixed results:



This is because our algorithm is a **hybrid** algorithm, and required classical optimization between steps!

classiq Studio for a Hybrid Quantum Algorithm

We were able to use the *classiq* studio workspace to run our classical optimizations!

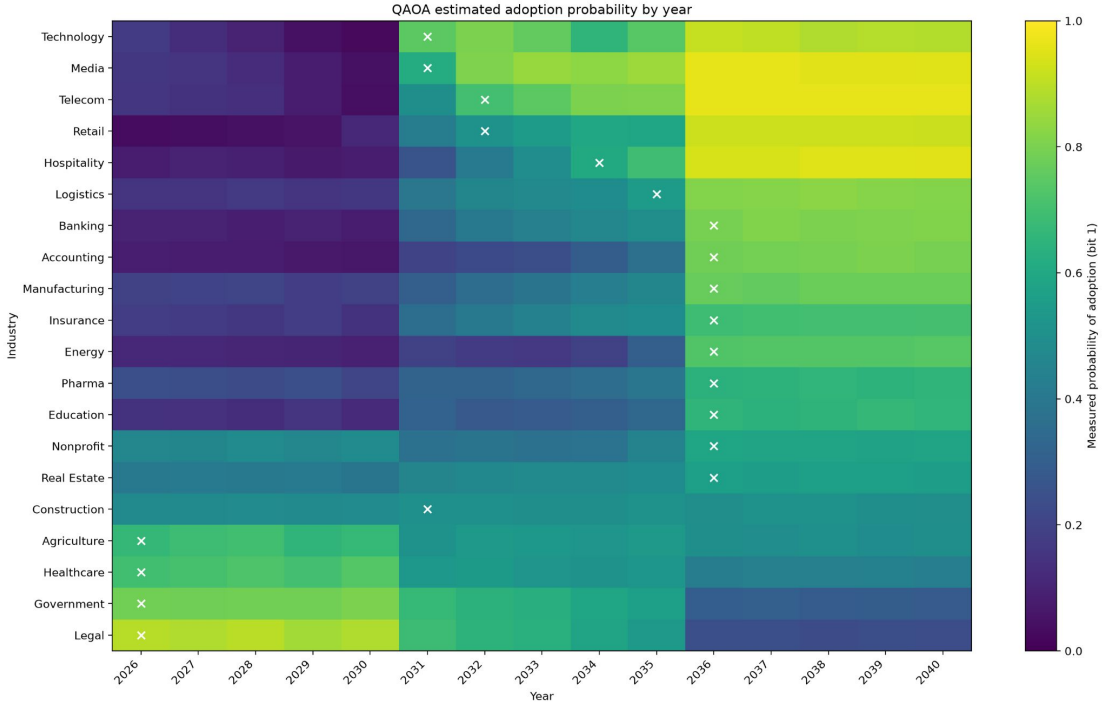


The workspace provided a great framework for us to implement our classical functions.

This allowed us to **link our script to our quantum circuit** using the circuit ID, and get some optimized results!

Getting Some Results with *classiq*!

Using this workflow, we were able to **produce results** using *classiq* simulated qubits with regards to solving the *Vasion* problem!



Shown left is a heatmap of the flip years for the various base industries.

Some results were strange, such as the bottom four industries flipping early, but the rest looks **fairly close to what we had predicted classically**.